SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

CO1-ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA0311	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
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Section / Batch:	AIML	**Date:**	10-10-2026

Code of Conduct

I P. Sahith Sai (192425029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student : P. Sahith Sai

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Annexure A – Application Based Questions

Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name.
Identify email addresses.
Extract mobile numbers.
Detect technical skills (Python, Java, SQL, Machine Learning, NLP).
Extract years of experience.
Generate a structured summary of the candidate's profile.
Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

Search products using an exact keyword.
Perform prefix-based searches.
Perform suffix-based searches.
Search using partial keywords.
Perform case-insensitive searches.
Display all matching product names.
Generate a report showing the total number of matching products for each search.

Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

Validate the student register number.
Validate the institutional email address.
Validate the course code.
Validate the semester information.
Validate the student's mobile number.
Display appropriate validation messages for each field.
Generate a final registration status report indicating whether the registration is successful.

NLP ASSESSMENT-1

Application-Based Questions Using Python Regular Expressions

Resume Information Extraction • Product Search • University Registration Validation

Introduction

Regular Expressions (Regex) are patterns used to search, extract, and validate text. Python provides the re module for operations such as search(), findall(), fullmatch(), and IGNORECASE. This document presents three application-based solutions using Regex.

Note: The interface images in this document are illustrative website-style mockups for presentation. They are not screenshots of a deployed or running website. The Python programs below are the implementations.

Objectives

- Extract useful information from resume text.
- Search a product catalogue using exact, prefix, suffix, partial, and case-insensitive matching.
- Validate student registration fields and report whether the registration is successful.

Section 1: Resume Information Extraction

1. Problem Statement

A recruitment company receives resumes in text format. The system must extract the candidate's name, email address, mobile number, technical skills, and years of experience, generate a short profile summary, and identify candidates with at least two years of experience and Python skill.

2. Objective

To automate basic resume screening using Python Regular Expressions.

3. Algorithm

- Read the resume as plain text.
- Use Regex patterns to find email, phone number, experience, and supported technical skills.
- Infer a candidate name from the first non-empty line when it is not an email or heading.
- Create a short profile summary.
- Mark the candidate eligible if experience is at least two years and Python is among the detected skills.

4. Python Program

```
import re

resume = """
Alex Johnson
Email: alex.johnson@example.com
Phone: +1 415-555-0182
Experienced Software Engineer with 3 years of experience.
Skills: Python, Java, SQL, Machine Learning, NLP
"""

email_match = re.search(
    r"\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\b",
    resume
)
phone_match = re.search(
    r"(?<!\w)(?:\+?\d{1,3}[-.\s]?)?(?:\(?\d{3}\)?[-.\s]?)\d{3}[-.\s]?\d{4}\b",
    resume
)
experience_match =
    re.search( r"(\d+(?:\.\d+)?)\s*\+?\s*years?\s+(?:of\s+)?experien
    ce",  resume, re.IGNORECASE
)

lines = [line.strip() for line in resume.splitlines() if line.strip()]
name = lines[0] if lines and "@" not in lines[0] else "Not found"

 skill_patterns =
     {  "Python":
    r"\bPython\b",  "Java":
    r"\bJava\b",
    "SQL": r"\bSQL\b",
    "Machine Learning": r"\bMachine Learning\b",
    "NLP": r"\bNLP\b"
}
skills = [
    skill for skill, pattern in skill_patterns.items()
    if re.search(pattern, resume, re.IGNORECASE)
]

email = email_match.group() if email_match else "Not found"
phone = phone_match.group() if phone_match else "Not found"
years = float(experience_match.group(1)) if experience_match else 0

summary = (
    f"{name} has {years:g} years of experience and skills in "
    f"{', '.join(skills) if skills else 'no listed technical skills'}."
)
eligible = years >= 2 and "Python" in skills

print("Candidate Name:", name)
print("Email:", email)
print("Mobile:", phone)
print("Technical Skills:", ", ".join(skills) if skills else "None found")
print("Years of Experience:", years)
print("Profile Summary:", summary)
print("Eligibility:", "Eligible" if eligible else "Not eligible")
```

5. Sample Output

```
Candidate Name: Alex Johnson
Email: alex.johnson@example.com
Mobile: +1 415-555-0182
Technical Skills: Python, Java, SQL, Machine Learning, NLP
Years of Experience: 3
```

Profile Summary: Alex Johnson has 3 years of experience and skills in Python, Java, SQL, Machine Learning, NLP.
Eligibility: Eligible

6. Interface Illustration

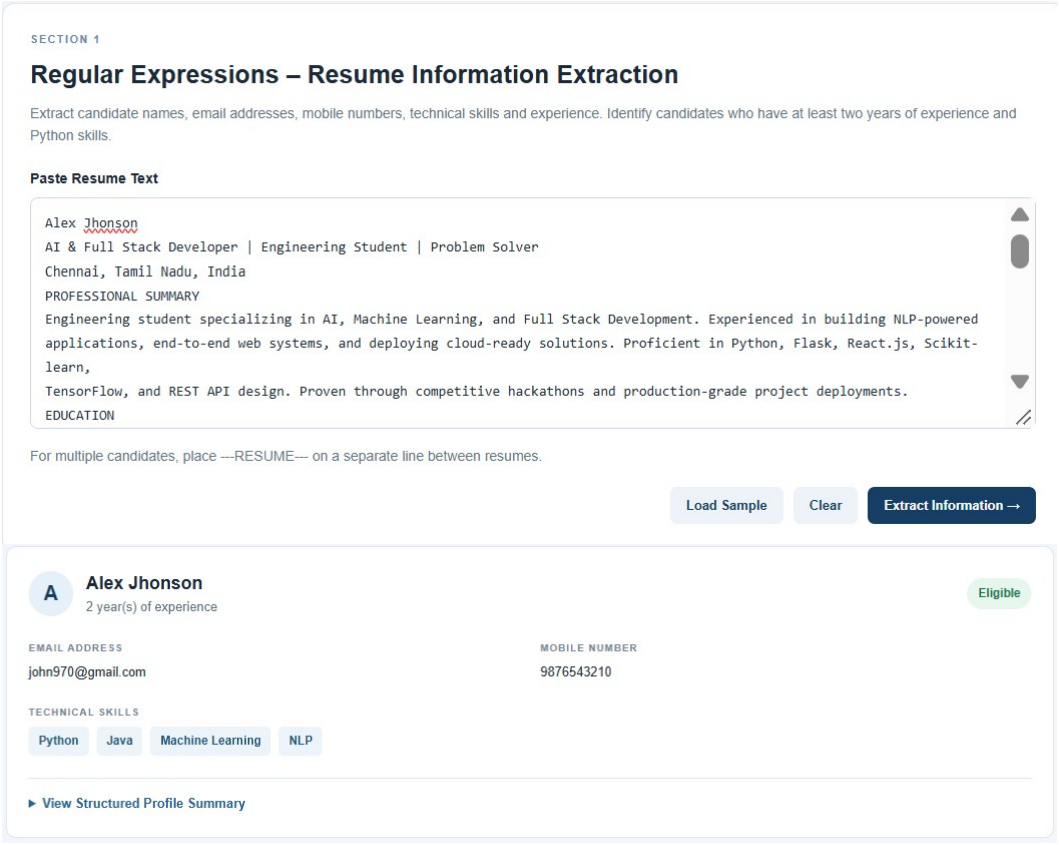


Figure 1: Illustrative Resume Information Extractor interface (mockup).

7. Result and Conclusion

The program extracts common resume fields and checks the minimum eligibility condition. Regex-based extraction depends on the format of the resume; a production system should handle varied layouts, international phone formats, and names more robustly.

Section 2: Product Search System

1. Problem Statement

An e-commerce catalogue needs flexible product search. The system must support exact, prefix, suffix, partial-keyword, and case-insensitive searches, display all matches, and report the number of matches.

2. Objective

To implement multiple keyword-matching modes using Regex.

3. Algorithm

- Store product names in a list.
- Read the keyword and selected search mode.
- Escape the keyword with re.escape() so special characters are treated literally.
- Build the corresponding Regex pattern for exact, prefix, suffix, or partial matching.
- Use re.IGNORECASE when case-insensitive matching is selected.
- Display the matching product names and total count.

4. Python Program

```
import re

products =
   [ "Laptop Pro
   14",   "Laptop
   Stand",
   "Wireless Mouse",
   "Gaming Laptop",
   "Laptop Backpack",
   "USB Keyboard",
   "Smartphone"
]

def search_products(keyword, mode="partial", case_insensitive=True):
   keyword = keyword.strip()
   if not keyword:
      return []

   escaped = re.escape(keyword)
   patterns = {
      "exact": rf"^{escaped}$",
      "prefix": rf"^{escaped}",
      "suffix": rf"{escaped}$",
      "partial": rf"{escaped}"
   }

   pattern = patterns.get(mode, patterns["partial"])
   flags = re.IGNORECASE if case_insensitive else 0
   return [product for product in products
         if re.search(pattern, product, flags)]

keyword = input("Enter product keyword: ")
mode = input("Choose exact/prefix/suffix/partial: ").strip().lower()
case_choice = input("Ignore case? (y/n): ").strip().lower()
ignore_case = case_choice == "y"
matches = search_products(keyword, mode, ignore_case)
```

```
print("\nMatching Products:")
if matches:
    print("-", item)
```

5. Sample Runs

```
Input:
Enter product keyword: laptop
Choose exact/prefix/suffix/partial: partial
Ignore case? (y/n): y

Matching Products:
- Laptop Pro 14
- Laptop Stand
- Gaming Laptop
- Laptop Backpack
Total matching products: 4
```

Examples of search modes: exact matches the entire product name; prefix matches names starting with the keyword; suffix matches names ending with the keyword; partial matches the keyword anywhere in the name.

6. Interface Illustration

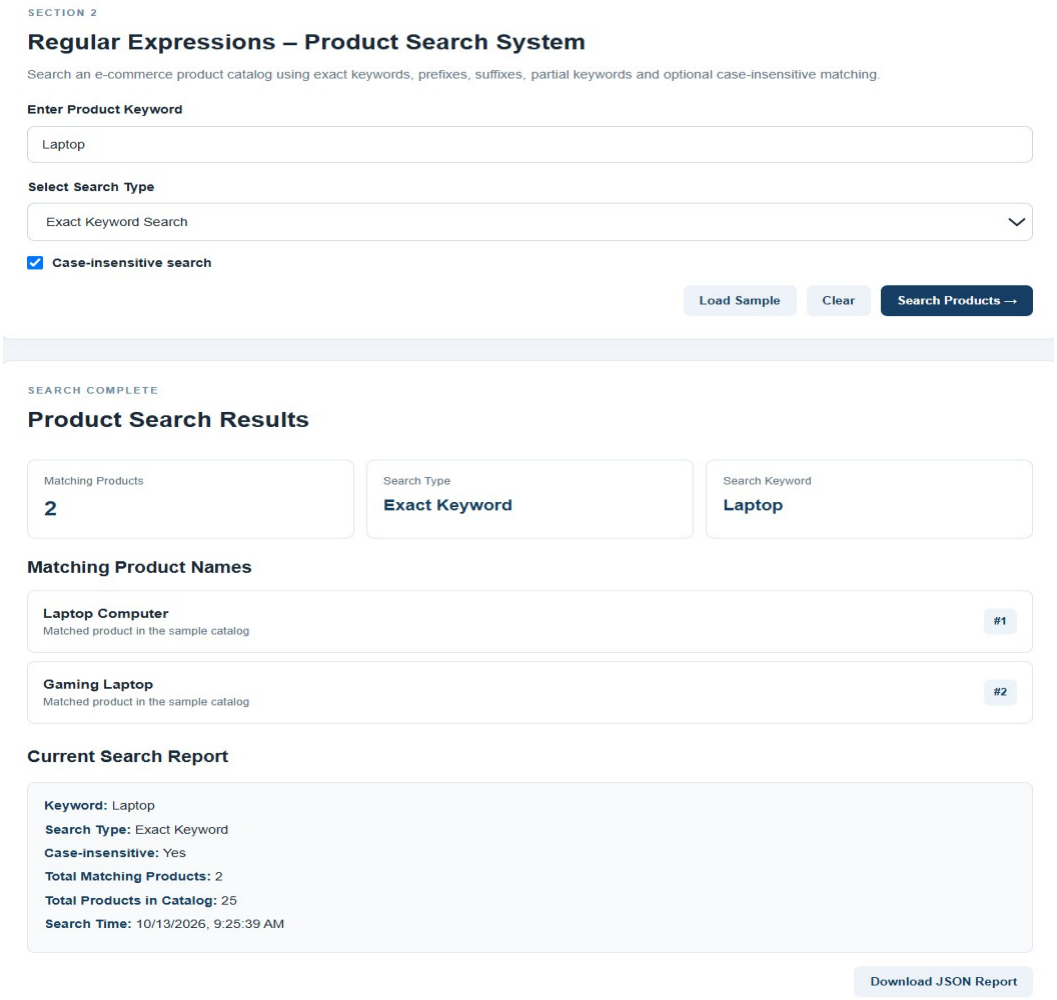


Figure 2: Illustrative Product Search System interface (mockup).

7. Result and Conclusion

The program supports four matching modes and optional case-insensitive matching. The report displays every matching product and the total count, helping users find products flexibly.

Section 3: University Registration Validation

1. Problem Statement

A university registration portal must validate the student register number, institutional email, course code, semester, and mobile number. It must display field-wise messages and produce a final registration status.

2. Objective

To validate registration fields with Regex patterns and simple range checks.

3. Validation Rules

- Register number: SIMATS + four-digit year + one letter + three digits (example: SIMATS2026A104).
- Institutional email: username followed by @saveetha.com.
- Course code: four letters followed by three digits (example: AIML301).
- Semester: integer from 1 to 8.
- Mobile number: 10 digits, beginning with 6, 7, 8, or 9.

4. Python Program

```
import re

register_no = input("Register number: ").strip()
email = input("Institutional email: ").strip()
course_code = input("Course code: ").strip().upper()
semester = input("Semester (1-8): ").strip()
mobile = input("Mobile number: ").strip()

checks = {
    "Register Number": bool(re.fullmatch(
        r"SIMATS\d{4}[A-Z]\d{3}", register_no, re.IGNORECASE
    )),
    "Institutional Email": bool(re.fullmatch(
        r"[A-Za-z0-9._%+-]+@saveetha\.com", email, re.IGNORECASE
    )),
    "Course Code":
        bool(re.fullmatch(  r"[A-
        Z]{4}\d{3}", course_code
    )),
    "Semester": bool(re.fullmatch(r"[1-8]", semester)),
    "Mobile Number": bool(re.fullmatch(r"[6-9]\d{9}", mobile))
}

print("\nValidation Report")
for field, valid in checks.items():
    print(f"{field}: {'Valid' if valid else 'Invalid'}")

if all(checks.values()):
    print("Final Status: Registration Successful")
else:
```

5. Sample Run

```
Register number: SIMATS2026A104
Institutional email: student@saveetha.com
Course code: AIML301
Semester (1-8): 5
Mobile number: 9876543210

Validation Report
Register Number: Valid
Institutional Email: Valid
```

Course Code: Valid
Semester: Valid
Mobile Number: Valid
Final Status: Registration Successful

6. Interface Illustration

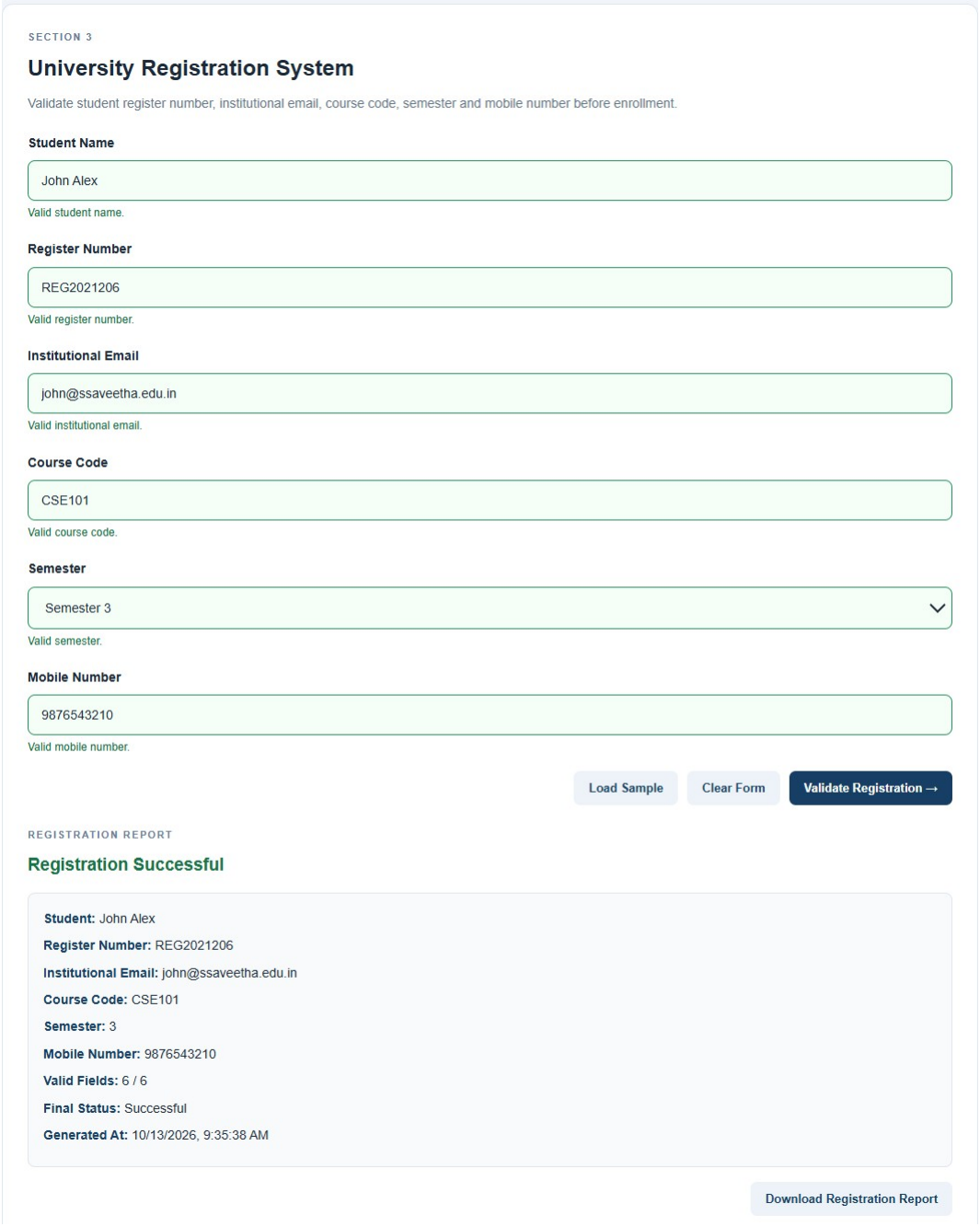


Figure 3: Illustrative Student Registration Validator interface (mockup).

7. Result and Conclusion

The program checks all five fields and reports each field's status. Registration is successful only when every validation passes. The register-number and course-code formats are examples based on the stated assumptions; they should be adjusted to match the university's official format.

Overall Conclusion

This assessment demonstrates three practical uses of Python Regular Expressions: information extraction, flexible product searching, and input validation. Patterns, re.search(), re.fullmatch(), re.IGNORECASE, and re.escape() help automate text-processing tasks. Correctness depends on clearly defined patterns and testing against valid and invalid examples.

References

Python Software Foundation. “re — Regular expression operations.” https://docs.python.org/3/library/re.html

Python Software Foundation. “Regular Expression HOWTO.” https://docs.python.org/3/howto/regex.html

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	Problem-Solving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____